

PRAYCG Control Center - Master Changelog

Scope and how to read this history

This record follows the retained software lineage from the v0.95a baseline through v1.0.0-alpha.10.0.4, oldest first. “Alpha” below abbreviates the v1.0.0-alpha version prefix. Entries describe changes documented in the corresponding retained package, not proposed features or promises that every earlier defect was eliminated.

The evidence comprises 42 numbered package identities, including v0.95a, and two separately named Alpha 9.9.0 catalog-repair archives. Same-version release notes, READMEs and version manifests take precedence over later summaries. Some early packages use their README as their release record. The Alpha 10.0.1-10.0.4 public changelogs supply their detailed current deltas. This document is a historical synthesis, not a replacement for the current installation guide or release-specific validation evidence.

Version gaps are not filled by inference. Alpha 9.1/9.2 and 9.7/9.8 appear as development scopes incorporated into later cumulative releases; separate packages were not found. No corresponding retained numbered artifact was found for Alpha 8.1, 8.5.0, 9.4, 9.5, 9.9.1 or 9.9.2. The two catalog-repair filenames do not establish releases 9.9.1 or 9.9.2. Internal component versions also differ legitimately from the Control Center package version.

v0.95a - Retained baseline and installation repair

v0.95a retained the existing native acquisition and analysis foundation while repairing installation compatibility. It bounded setup tools to accommodate BrainFlow's fallback dependency, explicitly required pyserial, and added hardware-free checks of native-library loading and package metadata. Installer success depended on those checks. Public version labels were updated while the v0.95 settings schema remained; token-bearing OSF links were removed from public documentation. Acquisition algorithms, native protocol sequences, analysis formulas, eligibility rules and frozen exploratory candidates were not changed by this hotfix.

Alpha 1 - Integrated platform foundation

The first integrated platform release added Protocol Forge, a reviewed declarative-module runner, Hardware Registry with canonical roles, an observer-first Acquisition Supervisor, and combined study/session locks. Analysis capability resolution, marker/timing contracts, BIDS export, privacy scanning and analysis-only environment specifications extended the platform. Cue-Locked Gamma Forensics remained exploratory. The Platform Alpha/Workbench interface sat alongside preserved PRAYCG3, PRAYCG4 and SMG structures. Reported software tests did not establish physical-device operation, interactive presentation timing, human feasibility, external BIDS conformance or container execution.

Alpha 2 - One console and one-use acquisition authority

Parallel workbench surfaces were consolidated into an operator console. Runtime locks bound the exact protocol, runner, engine and media to a fresh run identifier and unused output directory. A shared-log-root owner and process-bound, one-use lease governed acquisition, with heartbeats, interrupted-run finalization and ordered downstream evidence. Unrelated child processes could not inherit acquisition authority. EEG-plus-ALS accounting was corrected to four actual outlets, and stream health was separated from the controlled optical-placement check. A selected protocol or approved profile still did not itself authorize acquisition.

Alpha 3 - Runnable Protocol Atlas

Eleven Atlas adaptations joined three native protocols, yielding 14 entries. Initial families covered rest, narrative audio, repeated viewing, event segmentation, visual oddball, motor imagery, repeated clips and sequential visual narrative. Individual wrappers used a shared declarative engine with asset contracts, response records and analysis handoffs. Deterministic, asset-aware compilation, recursive validation and exact external-asset hashes strengthened locks. Discovery remained confined to packaged roots; reviewed defaults governed parameters and interrupted Atlas acquisition could not resume. Software flip markers were not physical photon-onset measurements.

Alpha 4 - Stimulus, hardware and analysis catalogs

Stimulus Forge and Library introduced typed packs, including four CC0 demonstration packs covering the 14 protocols. The hardware foundation cataloged eight adapters and fifteen routes, with a generic serial/UDP EDA/PPG prototype. Ten analysis modules received discovery/readiness metadata. Supplied third-party bridge archives contributed metadata rather than copied source. Demonstration material remained `DEMO_ONLY`; catalog inclusion, synthetic transport results and discoverability did not approve physical hardware, participant collection or scientific endpoints.

Alpha 5 - Connected preparation workflow

Seven work areas connected protocol selection, compatible stimuli, reviewed hardware, running and analysis. Locking a protocol filtered preparation choices; incompatible changes cleared or invalidated dependent selections. Governed master-video intake included an explicit checked, backed-up migration of older master-media tags. Hardware used a validated tray, reviewed profile and separate interface lock, with bridge actions revalidated when clicked. Protocol AI Docs Builder produced hashed, non-executable design packets. A design document could not install a runner or acquire runtime authority.

Alpha 6 - Persistent studies and governed analysis

Study Workspace persisted the progression from protocol and materials through hardware, acquisition, QC, analysis, review and export. Upstream changes preserved evidence while invalidating dependent approvals. Eleven execution modules gained registry-bound orchestration, eligibility checks, dependency planning, immutable pre-outcome locks and separately labeled post-lock exploratory additions. Local results bundles combined allowlisted derived outputs, redacted provenance, review records and checksums. Raw biosignals were excluded from that export path; local bundle creation neither uploaded the study nor granted publication approval.

Alpha 6.1 - Mature Endpoint Atlas integration

Seventeen additional PRAYCG runners expanded the catalog to 31 entries: 28 Atlas and three native. Explicit endpoint contracts covered manipulation, timing, signal quality, artifacts, respiration, order and anti-circularity. Compound trials gained step-level markers, trial-addressable responses and response summaries. A Mature Endpoint Evaluator and templates audited structural evidence and claim limits. They did not calculate physiological endpoints or validate the underlying theories. This was a distinct integration release over the separately preserved Alpha 6 baseline, not merely a renamed component.

Alpha 7 - Guided preparation and explicit assets

Plain-language areas introduced Home, Choose Protocol, Prepare Materials, Connect & Check Equipment, Run Session, Review Results and Settings. Named file/folder pickers replaced guessing multiple asset roles from a directory. Session confirmation froze selected paths and hashes. Runner-generated protocols explicitly skipped unnecessary file selection, while native protocols retained prepared-stimulus selection. The 31-protocol inventory, persistent workspace, governed analysis and local results bundles carried forward; simplifying navigation did not change approval or acquisition authority.

Alpha 8 - Public-study adaptations and live equipment visibility

Twelve adaptations expanded Atlas to 40 protocols plus three native entries. Added families included ERP CORE tasks, inner speech, meditation and SSVEP. Balanced trials were materialized for the participant/session before locking. Image-based tasks required reviewed manifests and hashes; SSVEP bound refresh/display parameters and recorded flip evidence. Six other imports remained catalog-only. A live monitor exposed stream quality and physiology summaries, with visible equipment-session identity and preflight progress. Adaptations were not exact replications; display timing, rights and human feasibility still needed separate evidence.

Alpha 8.2 - Numeric transport and bounded bench testing

EEG, auxiliary and ALS chunks were explicitly flushed with sample timestamps, addressing outlets that could be discovered but delivered no numeric data. Cyton+Daisy counter progression and 125 Hz reconstruction were corrected, and duplicate bridge launches were refused. Initial Bench Test Mode allowed only a narrowly evidenced EEG-contact waiver, with a reason, reserved pseudonym and permanent nonparticipant labels. Stale evidence could not regain PASS, mode changes disarmed acquisition, and interrupted finalization could not restore ARM.

Alpha 8.3 - Memory and monitor stability

The live inlet was opened once rather than twice, bringing opening failures inside the error boundary. Reduced buffering, bounded queues and logs, in-place monitor rows, callback cancellation and streamed reports limited growth. Release validation ran sequentially with lower memory pressure. The recorded same-host measurements supported a software resource-use repair, not a general guarantee of memory consumption, hardware performance or uninterrupted long-duration acquisition.

Alpha 8.4 - Stable hardware drafts and clearer readiness

A stable mutable hardware-profile draft replaced recursively growing timestamped paths. Protocol, AI-document and prospective-builder controls were reorganized, and six explicit run-session rows clarified readiness. A Cerelog custom BrainFlow raw/LSL route was recorded as quarantined and physically untested. Missing custom-board support and required electrical-safety acknowledgements remained visible. Adding the route to the library did not authorize ARM or certify human-connected safety.

Alpha 8.4.1 - Approval-path repair

Profile approval repaired previously saved recursive paths before path-resolution operations encountered Windows length limits. Immutable approved-profile history used a fixed base, one timestamp and a bounded collision counter. Error reporting exposed the draft and target involved. This repaired the approval workflow without changing its review requirements or converting a draft profile into evidence of physical validation.

Alpha 8.4.2 - ALS bridge lifecycle checks

Surviving packaged bridges on a selected serial port were discovered using an exact script allowlist. Operator-approved stop/restart actions revalidated process identity and full command details; current-session reuse blocked duplicates. Barcode diagnostics distinguished session mismatch, ambiguity, no stream and inlet errors from a genuine no-data condition. Consumers did not take ownership of the board. The subsequent Alpha 8.5.1 release addressed a remaining Windows termination failure in this path.

Alpha 8.5.1 - Guarded Windows bridge replacement

After consent and exact identity revalidation, bridge replacement used Windows process-tree termination where non-force stopping had failed. Barcode setup handled current reuse, stale sessions and EEG-only replacement, while ambiguous ports, failed termination and absent serial devices blocked progress. Bounded discovery tolerated lingering old stream advertisements. Bench display-only barcode operation needed no dongle or stream, but could not supply physical placement evidence. A successful process-ownership repair did not itself establish a working replacement device.

Alpha 8.5.2 - Explicit nonparticipant readiness bypass

Bench testing expanded beyond the earlier contact-only waiver. Reviewed hardware definitions, protocols and materials could proceed through confirmation and ARM despite listed readiness failures. A hash-bound artifact recorded the bypassed checks, reason and nonparticipant identity; restart or setup/mode changes revoked it. This enabled workflow testing with absent or unsuitable equipment without representing the missing checks as physical PASS or granting participant eligibility.

Alpha 8.5.3 - Bench role handling and terminal ownership

The Atlas hardware-role gate honored the fully bound bench bypass and recorded omitted roles in run evidence. A terminal process known to be dead no longer remained owner merely because its heartbeat was recent. Acquisition identity stayed immutable after lease/acquisition, requiring a new identifier for the next run. A narrow recovery repaired a specific older terminal-lease/replacement-lock inconsistency while preserving records; uncertain ownership remained blocked.

Alpha 8.5.4 - Windows stale-process detection

Windows process-existence checks replaced a problematic signal-style probe in both core compatibility layers. Invalid stale process identifiers were classified as dead instead of causing an interface exception. Access-denied or otherwise uncertain identities still blocked authority release. The change enabled the existing bounded recovery logic; it neither killed arbitrary processes nor supplied new participant or hardware validation.

Alpha 9.0.0 - Hardware Forge and stream contracts

Hardware Forge, Doctor and Existing-LSL intake organized eleven adapter families and fourteen routes. Canonical stream contracts recorded exact identity, channels, source policy, rates and timing metadata. Missing or ambiguous matches blocked progress. Longer observations exposed rate drift, jitter, gaps, clock regression, disappearance and identity changes rather than hiding them through reconnection. Existing-LSL intake produced drafts. A bench-verified Polar RR contract did not constitute human-connected approval, and cataloged external bridges or SDKs were not automatically bundled or trusted.

Alpha 9.3.0 - Cumulative hardening and operator confidence

This cumulative release incorporated the documented 9.1 hardening, 9.2 admission and 9.3 operator-confidence scopes. It added local redacted diagnostics, actionable readiness states, stress/recovery simulations, copy-only migration, AdapterKit and sequential evidence-based hardware admission. Timing-calibration summaries, synthetic rehearsal, checklists and an evidence timeline supported operators. Accelerated two-, eight- and twenty-four-hour stress presets were simulations, not actual wall-clock soaks. Diagnostic bundles excluded raw signals, and rehearsal or catalog evidence could not grant participant ARM.

Alpha 9.3.1 - Cancellation and archived-run recovery

Cancelling initial runner configuration wrote an immutable pre-start cancellation record before logger, marker or presentation startup. Exact acquisition ownership could then finalize intentionally. Recovery of older cancellation cases required a narrow evidence shape, absence of acquisition artifacts and a fresh identifier. Manual finalization used the archived run's own evidence even after current setup changed. Normal return no longer appeared as an aborted exception; live, mismatched or uncertain runs remained protected.

Alpha 9.6.0 - Persistent analysis plans and research candidates

Session-bound analysis selections persisted with registry and dependency fingerprints. Execution views came from immutable plans, with managed receipts binding code, configuration, environment, inputs, commands and reviewed outputs. New sessions archived earlier state. ResearchBundleCandidate 0.2 described environment, replay policy and relationships such as reanalysis, correction and supersession. Offline verification checked hashes, plan binding, policy and local-path leakage. These remained local candidates; GUI/hardware replay and byte-equivalence were not established merely by producing a descriptor.

Alpha 9.6.1 - Scoped meditation analysis

A meditation-versus-thinking adapter required the exact protocol, one unambiguous XDF, one logical event set, EEG and the expected blocks. Artifact retention, band summaries and pooled/paired contrasts were explicitly descriptive, within-person and exploratory. Matching JSON/CSV event mirrors counted as one set; distinct competing event sets remained ambiguous. Registry entry points and path arguments were validated without shell execution. General QC and compatible modules remained visible with precise reasons when ineligible.

Alpha 9.9.0 - Cumulative Analysis Forge and EEG methods

The cumulative 9.7 governance, 9.8 EEG-methods and 9.9 hardening scopes introduced declared capability contracts and outcome-blind frozen recipes. Search, recommendations, dependency previews, templates and input fingerprints improved planning. Nine governed EEG workers covered quality, preprocessing, spectral screening, event-related analysis, connectivity, complexity, microstates, motor imagery and sensor-level representation. Missingness, minimum data and deterministic methods remained explicit. Screening did not become a canonical fitted model or a clinical endpoint. The no-network TorchEEG sandbox checked readiness, not trained-model accuracy.

Alpha 9.9.0 catalog-repair reissues

Two retained archives are labeled CATALOG_FIX and CATALOG_FIX_V2. Both retain the internal Alpha 9.9.0 package identity and release date. They are catalog-repair reissues, not evidence of separately numbered Alpha 9.9.1 and 9.9.2 releases. Their inherited notes do not provide a reliable separate patch narrative, so no finer change attribution is asserted here. The explicitly numbered Alpha 9.9.3 notes document the subsequent catalog repairs.

Alpha 9.9.3 - Catalog visibility and event discovery

A missing runtime import that emptied the analysis catalog was corrected. Old registry paths could be rebased, split XDF and runner-event evidence discovered, and equivalent event mirrors coalesced. An actionable default view retained diagnostics and clearer selection guidance. These repairs restored access to eligible modules without loosening input hashes, immutable locks or execution-time eligibility.

Alpha 9.9.4 - Truthful managed-worker outcomes

QC and meditation workers completing with cautions no longer appeared as process failures. Successful exit awaited review, expected ineligibility had a distinct outcome, and unexpected failures remained failures. Changed worker, descriptor or configuration fingerprints invalidated reuse of older locked plans. Operators needed a fresh plan rather than silent rewriting of scientific intent or historical execution evidence.

Alpha 9.9.5 - Drift-aware readiness and bounded recovery

Readiness retained raw variation while using a detrended measure to distinguish slow DC drift from residual noise. Flatness, disconnection, excessive residual noise, railing and nonfinite samples remained hard flags; drift stayed visible and did not prove electrode contact or usable EEG. ALS placement profiles reached native dialogs with their hashes. Narrow recovery addressed failed pre-start runs only when exact archived ownership and absence of acquisition evidence supported it.

Alpha 9.9.6 - Prospective runner configuration

Supported baseline, washout, final-baseline, display-fit and EEG-watchdog settings moved before session confirmation. Noncanonical choices required a reason and were frozen with their fingerprint. A ten-second test preset made shortened timing explicit. The runner inherited locked values instead of offering edits that would later fail. Strict field/type/bounds checks rejected malformed declarations. A shortened baseline remained a declared variant, not a scientifically equivalent canonical run.

Alpha 9.9.7 - Exact run handoff and media cleanup

Prepare Run for Analysis joined runner evidence to XDF using the governed run identifier rather than filename recency. Composite manifests distinguished complete and partial runs; required QC could inspect partial evidence without enabling complete-protocol conclusions. Changed inputs invalidated locked execution. Media cleanup used pause/unload handling and preserved diagnostics. Managed autonomic extraction, explicit channel selection, NumPy integration compatibility and a software-only 53-definition certification matrix broadened reliability checks without certifying codecs, devices or human sessions.

Alpha 9.9.8 - Marker-bounded analysis and participant warnings

Authenticated protocol start/end markers defined analysis windows separately from immutable raw recording containers. QC distinguished protocol quality from recorder tails; preprocessing and downstream analyses respected bound intervals. Candidate selection distinguished complete recordings from pre-start fragments while retaining ambiguity checks. Prominent participant/bench banners, additional confirmation for cautions or noncanonical settings, and a stop-recorder reminder clarified operator choices. Run dispositions could be downgraded additively, never rewritten into clean canonical evidence after the fact.

Alpha 9.9.9 - Authoritative EEG identity

Stream type and canonical role took precedence over vendor-name guesses. OpenBCI auxiliary and status streams could no longer overwrite actual EEG rate evidence. Resolution became independent of inventory order; invalid rates and multiple authoritative EEG streams remained unresolved. Core recommendations exposed exact eligibility errors and refreshed them for the selected run. This repaired classification, not the underlying signal: contamination, diagnostic status and existing run limitations remained unchanged.

Alpha 9.9.10 - Analysis execution and provenance repair

The locked dependency graph advanced one eligible node at a time, preserved terminal outcomes and allowed independent branches to continue. Autonomic configuration paths were resolved and hash-validated correctly. Missing required CAI/SID inputs yielded `NOT_ESTIMABLE`, not substitutes or zeros. Valid receipts reconciled stale output pointers, upstream cautions propagated, and downstream tools found authenticated Master Suite outputs. Verification-only preparation preserved locked manifests; post-hoc supplements retained separate exploratory authority.

Alpha 9.9.11 - Prospective dependencies and persistent results

Planning distinguished immediately runnable modules from those whose prerequisites could be created by the plan. Exact dependencies were added automatically, while unavailable recorded evidence still blocked execution. Deterministic pre-outcome recipes counted principal onset markers rather than incidental event rows. Persistent Results reconstructed exact-plan receipts and exposed missing, altered or corrupt evidence. Standardized summaries linked native reports without pretending to be scientific results; historical sessions were read-only and output paths confined to the workspace.

Alpha 9.10.0 - Unified Forge, behavioral modules and passive monitoring

One Analysis Forge combined QC, planning, execution and results. Authenticated required QC preceded module selection; cumulative immutable revisions preserved earlier decisions and post-hoc labels. Seven noninteractive legacy adapters improved managed analysis, with full-run visualization using the shared timeline. Fourteen exploratory behavioral modules brought the main library to 54 Atlas plus three native protocols. Optional passive MNE-LSL monitoring added traces, provisional diagnostics, simulation and replay. Closed-loop feedback and source reconstruction were not enabled; short default tasks were not power-validated designs.

Alpha 9.10.1 - Accessible reports and private study handoff

Shared input inventories restored supporting-record resolution, recipe review, explicit overrides, BIDS access and interrupted-run recovery. Results preferred substantive reports, added deterministic plain-English explanations and a readable master index, and exposed Full Chain child outcomes. Optional full-study/AI-ready exports inventoried available raw and supporting evidence with purpose and privacy caveats. AI protocol drafts could be reopened and external declarative candidates reviewed/staged. Packages were local, not automatically uploaded, de-identified, executed or admitted as runnable studies.

Alpha 9.10.2 - Session-safe planning and multi-stream replay

New-session handling stopped stale Forge revisions from reattaching old locks. Ownership and input/code bindings governed reuse; recovery preserved prior evidence and an exact-byte backup. Viewing outcomes could not become prospective simply by starting another session. Replay gained transport controls, seek, speed and a shared corrected timestamp axis across supported streams. Recorded hardware profiles were verified against the recording; current equipment could not substitute for missing historical context. Replay created no outlets, recordings or acquisition authority.

Alpha 10.0.0 - Reproducible Research Exchange foundation

Research Bundle 1.0 unified private replay, AI-ready review and results-review exports with a file ledger, dataset identity, environment descriptors and lineage. Explicit privacy selections governed raw/private inclusion. Verified import preserved originals separately from new work; archived code remained inert. Current-installed-build reanalysis froze inputs, source-plan selection and comparison tolerances, ran required QC and recorded new authority. Numeric comparisons disclosed coverage rather than asserting complete equivalence. The local DATA catalog did not upload, install dependencies or demonstrate independent second-machine reproduction.

Alpha 10.0.1 - Protocol-aware setup and cleaner distribution

Protocol-specific pre-start forms replaced universal controls: editable settings matched supported runner inputs and fixed designs stayed read-only. Drafts were isolated by protocol and hardware identity, and changes required reconfirmation. Seventeen neutral public titles described tasks or measurements while stable identifiers and former-name aliases preserved history. A shared Python 3.11 installer supported Core and isolated Live Monitor components. The public layout became five top-level files plus an application folder, with one current workflow guide and changelog. Short demonstrations remained distinct from Bench Test Mode.

Alpha 10.0.2 - Completed sessions and reused process identifiers

Protocol changes and configuration edits stopped treating cached acquisition state as proof of a live run after exact archived evidence showed finalization. Only verified inactive terminal sessions could refresh the interface cache; history stayed immutable and ARM was not restored. A bounded Windows creation-time fallback distinguished some reused process identifiers. Unavailable, malformed, matching or insufficiently precise identity evidence continued to block release of authority. Old heartbeats alone did not justify killing a process or bypassing recovery.

Alpha 10.0.3 - Portable identity, profile-driven checks and synthetic QA

Explicit release metadata replaced installation-folder names in catalog consistency checks. Equipment checks and native watchdog settings followed the reviewed EEG profile rather than OpenBCI defaults. No-ALS profiles avoided unnecessary optical capture; required ALS was not waived. Native electrode-map generation was restricted to matching routes. Certification checked substantive schedules instead of empty-timeline success. Seeded XDF fixtures exercised actual QC, preprocessing and spectral workers; missing-time testing exposed and blocked filtering across gaps. These fixtures did not validate real headsets, unresolved Athena support, physical timing or exhaustive hardware-by-protocol execution.

Alpha 10.0.4 - Attribution and bounded AI-authoring continuity

Structured credits covered all 67 protocol definitions, separating Hoyt Banks's PRAYCG development/adaptation work from external source authors. Sixteen verified references covered 23 externally sourced protocols; historical source hashes and study identities remained preserved. Readable credits, registries and generation hooks retained attribution, while licensing guidance separated original PRAYCG code from third-party obligations.

Protocol AI returns were explicitly review-and-stage candidates, not automatically installed runners. Parent-linked, data-only AI recipe proposals required normal review and locking. Recipe revision preserved unexposed condition subsets and RSA matrices or required explicit resolution of incompatible edits. The shared fourteen-protocol behavioral runner accepted the environment-based manifest handoff while retaining package, runtime-lock and lease checks. No new scientific method, arbitrary executable extension admission, automatic cloud access or historical-study rewrite was introduced.

Boundaries that persist across the lineage

Software completion, byte integrity, catalog membership, a readable report and scientific validity remain different claims. Retained tests describe their particular software, synthetic or same-machine scope; they do not by themselves establish safe human-connected hardware, display/photodiode timing, clinical meaning, physiological origin, construct validity, causal interpretation, power or population generalization.

Original recordings, immutable locks and prior results remain provenance, including failed, incomplete, contaminated, noncanonical and non-estimable outcomes. Repairs must not retrospectively improve their classification. New analyses, summaries and revisions require their own evidence and honest prospective/post-hoc labels. Public titles and source citations do not grant endorsement, replication equivalence or rights to redistribute external materials. Local exports still require privacy, consent and licensing review before sharing. Consult the current guide and the exact release's validation artifacts for operational instructions and executed coverage.